Stanislav Smirnov

Cologne, Germany | +49 176 4345 8829 | mr.stassmirnoff@hotmail.com | linkedin.com/in/smirnov-stanislav | github.com/mrStasSmirnoff

Summary

Senior Data Scientist in payments/fintech focused on payment orchestration, routing, retries and decline recovery. Product-adjacent: I drive prioritization and stakeholder alignment, translate insights into shippable changes, and own KPI definition and post-launch evaluation.

Highlights

- **4x** uplift in fallback routing success and **3x** increase in recovered revenue by analyzing PSP performance on NA traffic and redesigning a low-performing rerouting path.
- **25–50%** improvement in retry success rates for in-scope transactions through the **Dynamic Retries** ML solution (idea → PoC → production rollout).
- Drove decision memos / rollout plans with PM+EM, defined success metrics, and owned launch evaluation and KPI tracking for routing and retry changes.

Skills

Languages: Python, SQL, PySpark
ML/AI: classical ML (Regression, SVM, Decision Trees, Boosting), Deep Learning (CNNs, LSTMs), LLMs
Payments: authorization/decline analysis, PSP performance, routing, retries, tokenization (network tokens)
Product: decision memos / PRDs, roadmap/backlog prioritization, KPI definition, stakeholder alignment, rollout evaluation
Cloud: AWS, Azure
Data Engineering: data warehouse design (Data Vault 2.0), ETL development
Tools: Git, Docker, Airflow, FastAPI, SageMaker, MWAA, Databricks, Azure Data Factory, TensorFlow, Keras

Experience

Senior Data Scientist, Payments — April 2024 – Present
Cleverbridge AG — *Cologne, DE*

- Improved fallback re-routing authorization success by **4x** (from 2–3% to 11%) by analyzing PSP performance on NA traffic and proposing a switch from a low-performing route to a higher-performing alternative; led to a **3x** increase in recovered revenue on that traffic slice.
- Owned end-to-end optimization lifecycle: opportunity discovery → decision memo → engineering alignment → rollout plan; defined KPIs and reviewed post-launch outcomes.
- Own the evolution of the **Dynamic Retries** ML solution as part of the payments stack, extending coverage to additional decline categories and use cases.
- Performed deep-dive analysis on BIN-level and regional authorization patterns across PSPs to identify weak routes and opportunities for optimization.
- Derive analytics on tokenization and network-token performance (coverage, success rates, impact on authorization) to inform routing and retry strategies.

Data Scientist — December 2021 – April 2024
Cleverbridge AG — *Cologne, DE*

- Originated and shipped **Dynamic Retries** (ML-driven retry timing optimization): wrote the proposal/PRD, aligned stakeholders, and led cross-functional delivery from PoC → MVP → production rollout.
- Achieved **25–50%** higher success rate for in-scope retries; owned KPI definition, measurement design, and rollout evaluation.
- Built a hybrid retry strategy combining a rule-based scheduling heuristic (country/currency × day-of-month × hour-of-day) with ML timing optimization to select high-conversion retry windows.
- Implemented and operated the solution on **AWS** (Python microservices, SageMaker, MWAA), including training/inference workflows and automation for reliable production runs.
- Designed the ML framework behind **CleverAutomations** and delivered churn-reduction models and automated workflows that identified at-risk customers and triggered targeted campaigns.

Data Engineer September 2019 – December 2021
Cleverbridge AG *Cologne, DE*

- Led the design and implementation of a scalable data warehouse on Azure (Data Vault 2.0), enabling reliable reporting and self-service analytics across the company.
- Built and optimized ETL pipelines (Azure Data Factory, Databricks, T-SQL) processing millions of records per day from multiple source systems.
- Managed schema changes and migrations with Flyway, ensuring consistent, versioned deployments across environments.
- Developed Python microservices and Airflow workflows for data products, containerized with Docker for resilient, maintainable operation.

Machine Learning Intern May 2017 – July 2017
KOSTAL Group *Dortmund, DE*

- Trained a gesture-recognition neural network using Keras and TensorFlow; achieved **80%** validation accuracy on top-5 gestures using static image inputs with a CNN.
- Improved model robustness through data preprocessing and augmentation.

Network Optimization Engineer May 2012 – March 2015
MegaFon *Republic of Bashkortostan, Russia*

- Led the **network optimization team** of three engineers.
- Reduced CS call drop rate in the 2G network by **0.2%**.
- Increased Inter-RAT handover success rate in the UMTS network by **0.8%**, improving overall network performance.

Projects

Object Detection in Fine Art Paintings (MSc Thesis) | *Python, Deep Learning, CNN* 2017 – 2018

- Developed a custom CNN-based approach for object detection in fine art paintings, improving detection accuracy by **18%** over existing methods.
- Built the training and evaluation workflow and performed systematic error analysis to iterate on model design.

Education

University of Paderborn Paderborn, DE
Master of Science in Electrical Systems Engineering (Signal Processing) *2015 – 2018*

- **Thesis:** Enhancing Object Detection in Fine Art Paintings Using Deep Learning Techniques (**+18%** detection accuracy vs. baseline).
- **Research Assistant:** statistical image processing; plasmonic nanoparticles modeling in COMSOL; Lie groups programming in GAP.

Publications & Certifications

Deep Learning for Object Detection in Fine Art Paintings | *IEEE Xplore* 2020
Nonlinear Dielectric Properties of Random Paraelectric-Dielectric Composites | *Acta Materialia* 2021
Dynamic Retries: failed payment recovery | *Technical writing*
AWS Certified Cloud Practitioner

Languages

Russian (Native) | English (Fluent) | German (Intermediate, B2)

Additional Information

Volunteer: Bashkortostan branch of the Russian Union of Young Scientists.
Awards: Appreciation letter from the CEO of the Republic of Bashkortostan branch of PJSC “MegaFon” for outstanding performance.